*.dtl

Purpose

A decision table is a set of rules. Each day SWAT+ checks the table's conditions for an object (an HRU, a reservoir, a water allocation object), finds which of the table's alternatives are satisfied, and performs the actions that those alternatives switch on. Decision tables drive automatic management (planting, harvest, fertilizer, irrigation), reservoir and paddy releases, land use change scenarios, and water allocation demands. See decision tables for the concept.

There are four decision table files. Their names are given on the `decision_table` line of `file.cio`, in this order:

File	Default name	Used by
land use and management	`lum.dtl`	HRUs, through the automatic operations listed in `management.sch`; HRU-lte growing seasons; manure demands in `manure_allo.mnu`; irrigation demands, outside-source concentrations and treatment in water allocation
reservoir release	`res_rel.dtl`	reservoirs and HRU wetlands or paddies that name a release table (see `*.res` and `*.wet`)
land use scenarios	`scen_lu.dtl`	tables listed in `scen_dtl.upd`, checked for every HRU every day
flow control	`flo_con.dtl`	saturated buffers, and water allocation demand and source or receiver fractions

A file that is missing, or named `null` in `file.cio`, gives an empty set of tables.

Format

All four files have the same layout. Free format, whitespace-separated.

- Line 1: title (skipped).
- Line 2: the number of decision tables in the file.
- Line 3: skipped (normally blank).
- Then each table in turn:
  - a header line (skipped);
  - the table line: `name`, `conds`, `alts`, `acts` (table name, number of conditions, number of alternatives, number of actions);
  - a header line (skipped);
  - `conds` condition rows;
  - a header line (skipped);
  - `acts` action rows;
  - in `res_rel.dtl`, `scen_lu.dtl` and `flo_con.dtl` only: one more line, which is skipped. Leave a blank line here. Without it the next table's header is skipped in its place and the following lines are read out of step.

The counts on the table line must match the rows that follow. Extra or missing rows shift every later table. Names are matched exactly, and a name that is not found is not reported.

Condition rows

#	Column	Description
1	`var`	Condition variable (see Condition variables).
2	`obj`	Object type the variable belongs to, for example `hru`, `res`, `null`. For `prob_unif`, this column contains the application probability instead (see below).
3	`obj_num`	Object number. 0 means the object the table is being checked for.
4	`lim_var`	Limit variable. A name (land use, plant, soil group) or a reference volume, depending on the condition. `null` when not used.
5	`lim_op`	Limit operator: `*`, `+`, `-` or `/`. Combines `lim_var` with `lim_const` for `soil_water`, `vol` and `vol_wet`. `-` when not used.
6	`lim_const`	Limit constant.
7 and on	`alt1`, `alt2`, …	One alternative operator per alternative.

For most conditions the current value of the variable is compared with `lim_const`. For `soil_water`, `vol` and `vol_wet` it is compared with a target built from `lim_var`, `lim_op` and `lim_const`, for example "emergency volume × 0.8".

Alternatives

Each alternative is a column of operators, one per condition. An alternative is satisfied when every one of its conditions is true. For numeric conditions the operators are:

Operator	Condition is true when
`<`	value < limit
`>`	value > limit
`<=`	value limit
`>=`	value limit
`=`	value = limit
`/=`	value limit
`-`	always (the condition does not apply to this alternative)

Any other text in a numeric condition's alternative column is treated like `-`. In particular `/` does not mean "not equal" for numeric conditions.

The yes/no conditions (`plant_gro`, `plant_name_gro`, `hyd_soil_group`, `land_use`, `cal_group`, `tillage`, `prob_unif`, `prob_unif1`, `prob_unif_lu`, `ch_order`) use only `=`, meaning the condition must be true. `tile_drained` and `plant_com` also accept `/`, meaning the condition must be false. Every other operator is treated like `-`.

Action rows

#	Column	Description
1	`act_typ`	Action type (see Land use and management actions and Reservoir release actions).
2	`obj`	Object type. For `irrigate`, the type of the water source.
3	`obj_num`	Object number. 0 means the object the table is being checked for.
4	`name`	Action name. Used to link `days_act` conditions to this action, and by `irrigate`, `irr_demand` and `puddle` as described below. Otherwise free text.
5	`option`	Action option, specific to the action type.
6	`const`	First action constant.
7	`const2`	Second action constant. In `lum.dtl`, values below 1 are raised to 1.
8	`fp`	File pointer: a name in another input file, specific to the action type.
9 and on	`outcome1`, `outcome2`, …	`y` or `n` for each alternative.

An action runs when at least one alternative is satisfied and the action's outcome for that alternative is `y`. Actions run in the order of the rows.

In `lum.dtl`, many actions use `const2` as the maximum number of times the action may run in a calendar year for each HRU. The count is reset on 31 December. These actions are marked "per year: `const2`" in the table below.

Condition variables

Units are those of the compared value. "Plants" means the plants of the HRU's plant community.

var	Value compared	Notes
`w_stress`	Average water stress factor of the growing plants (1 = no stress)	1 when no plant is growing.
`n_stress`	Average nitrogen stress factor of the growing plants	0 when no plant is growing.
`p_stress`	Average phosphorus stress factor of the growing plants	0 when no plant is growing.
`phu_plant`	Fraction of potential heat units accumulated by the first plant that has started accumulating (plant 1 if none)	
`phu_base0`	Fraction of the annual base-zero heat units accumulated at the HRU's weather station	`obj` must be `hru` or `hlt`.
`precip_cur`	Precipitation today, mm	`obj` must be `hru` or `hlt`.
`precip_next`	Precipitation tomorrow, mm	`obj` must be `hru` or `hlt`.
`plant_gro`	Plant number `lim_const` (1 if 0) is growing	Yes/no.
`plant_name_gro`	The plant named in `lim_var` is in the community and growing	Yes/no.
`days_plant`	Days since the last planting	
`days_harv`	Days since the last harvest	
`days_irr`	Days since the last irrigation	
`days_act`	Days since the action named in `lim_var` (an action `name` in the same table) last ran	Compared with `lim_const` + 2.
`day_start`	Day of year on which the simulation started	
`irr_year`	Irrigation applied to the HRU, mm	In 62.0.1 the total is not reset each year, so it is the total since the start of the run.
`slope`	HRU slope, m/m	
`soil_water`	Soil water content above wilting point, mm	Target: `lim_var` (`wp` = 0, `fc` = water at field capacity, `ul` = water at saturation), combined with `lim_const` by `lim_op`.
`jday`	Day of year	
`month`	Month	
`year_rot`	Year of the crop rotation	Starts at the plant community's initial rotation year and increases by 1 each 31 December.
`year_gro`, `cur_yrs_mat`	Years of growth of plant 1	Same value for both names.
`year_cal`	Calendar year	
`year_seq`	Year of the simulation (1 for the first year)	
`year_start`	First calendar year of the simulation	
`biomass`	Above-ground plant biomass, kg/ha	
`leaf_area`	Leaf area index of all plants	
`ground_cov`	Above-ground plant biomass, kg/ha	Residue is not included in 62.0.1, so it gives the same value as `biomass`.
`p_factor`	USLE P factor	
`k_factor`	USLE K factor of the top soil layer	
`hyd_soil_group`	Soil hydrologic group equals `lim_var` (`A`, `B`, `C` or `D`)	Yes/no.
`p_pet`	Ratio of average annual precipitation to PET from the weather generator	
`p_lab_150`	Labile phosphorus in the upper soil, ppm	Uses the first two soil layers.
`soil_temp2`	Temperature of soil layer 2, °C	
`soil_carbon`	Organic carbon of soil layer 1, %	
`tile_drained`	The HRU has tile drains	`=` true, `/` false.
`prob`	A uniform random number between 0 and 1	Compared with `lim_const`; use `<` to act with probability `lim_const`.
`prob_unif`	Today is the drawn application day	See Probability conditions.
`prob_unif1`	Random draw within a window that starts on 1 January	See Probability conditions.

var	Value compared	Notes
prob_unif_lu	Random draw that spreads an operation over the HRUs of a land use	See Probability conditions.
channel_flo	Flow in the current hydrograph, m^3/s	
tile_flo	Tile flow from the HRU today, mm	
irr_demand	Irrigation demand of the HRU, m^3	
irr_pou_dmd	Demand of water allocation place of use obj_num, m^3	
aqu_dep	Depth to the water table of aquifer obj_num, m	
land_use	HRU land use equals lim_var (a name in landuse.lum)	Yes/no.
cal_group	HRU calibration group in landuse.lum equals lim_var	Yes/no.
tillage	The HRU's management schedule lists the decision table named in lim_var	Yes/no.
plant_com	The plant named in lim_var is in the plant community	= true, / false.
ch_order	Stream order of channel obj_num equals lim_const	Yes/no.
wtow_vol	Water in water tower obj_num, m^3	Target: lim_const × tower capacity.
vol	Reservoir volume, m^3	Target: lim_var pvol (principal volume) or evol (emergency volume) combined with lim_const by lim_op. With + and −, lim_const is in ha · m. With lim_var e-pv, the target is $V_p + \text{lim_const}\,(V_e - V_p)$.
res_inflo	Inflow to reservoir obj_num, m^3/s	
wet_depth	Water depth of the HRU wetland or paddy	lim_const in mm.
weirh	Weir height of the HRU wetland or paddy	lim_const in mm.
vol_wet	Water volume of the HRU wetland or paddy, m^3	Target: lim_var pvol or evol combined with lim_const by lim_op.

Probability conditions

These three conditions let an operation happen on a random day within a window instead of on a fixed date.

prob_unif uses the columns differently from other conditions: obj is the probability f that the operation happens at all this year, obj_num is the first day of the window and lim_const is the last day (day of year). On the first day of the window SWAT+ decides, with probability f, whether the operation will happen this year, and if so draws one day in the window at random. The condition is true only on that day. A window that ends before it starts (for example 330 to 30) runs over the turn of the year.

prob_unif1 uses obj_num as a calendar year and lim_const as a number of days. The window starts on 1 January of that year. On each day of the window the condition is true with probability $1/n$, where n is the number of days left in the window.

prob_unif_lu uses obj_num as the first day and lim_const as the last day of the window. It is meant to be combined with a land_use condition in the same table: it spreads the operation over the HRUs of that land use so that the share of HRUs treated by day d follows a uniform distribution over the window.

Land use and management actions

Actions in lum.dtl. Unless stated otherwise, obj is hru and obj_num 0 means the HRU being checked.

act_typ	option	const	const2	fp	Effect
plant	plant name in the community	-	-	name in transplant.ops	Starts growth of the plant if it is not already growing. With a transplant entry, the plant starts from the transplant state (see management.sch).
harvest	plant name, or all	residue harvest efficiency	per year	name in harv.ops	Harvests the plant.
harvest_kill	plant name, or all	residue harvest efficiency	per year	name in harv.ops	Harvests and kills the plant.
kill	plant name, or all	-	per year	-	Kills the plant; biomass becomes residue.
till, tillage	name in tillage.til	-	per year	-	Tillage.
fertilize	name in fertilizer.frt	amount, kg/ha	per year	name in chem_app.ops	Applies fertilizer. On an HRU with standing water and a method with surf_frac 1, the fertilizer goes into the water.
fert_future	name in fertilizer.frt	amount, kg/ha	day of year	name in chem_app.ops	Schedules a fertilizer application on the given day. obj_num is the sequence number of the fert_future action in the table.
manure	-	amount, kg/ha	per year	-	In 62.0.1 the manure and method names are not looked up, so this action applies an empty fertilizer entry. Use fertilize with a manure from fertilizer.frt, or manure_demand.
manure_demand	name in chem_app.ops	amount, t/ha	per year	-	Sets a manure request for manure allocation. obj_num is the manure source number.
pest_apply	pesticide name listed in constituents.cs	amount, kg/ha	per year	name in chem_app.ops	Applies the amount times the method's app_eff.
graze	name in graze.ops	-	-	-	One day of grazing.
burn	name in fire.ops	-	per year	-	Burns plants and residue.

act_typ	option	const	const2	fp	Effect
irrigate	name in irr.ops	depth, mm	per year	-	Irrigates with const · eff_frac · (1 − sumq_frac). obj and obj_num name the source: aqu, cha or sdc, or res takes the water from that object and irrigates nothing if it has too little; any other obj (for example hru) is an unlimited source. With name ponding, the HRU is a paddy: const is the target ponding depth (mm), const2 the depth below which water is added, and the depth applied is the shortfall from the target.
irr_demand	name in irr.ops	depth, mm	per year	unlim or other	Sets the HRU's irrigation demand for water allocation (const× area). With fp unlim, the amt_mm of the irr.ops entry is applied directly. With name ponding, const and const2 are the target and threshold ponding depths (mm).
res_irr_dmd	-	depth, mm	-	-	Records an irrigation demand on reservoir obj_num.
rot_reset	-	rotation year	-	-	Sets the rotation year (minimum 1).
plant_reset, harv_reset, kill_reset, irr_reset	-	days	-	-	Sets the days since the last planting, harvest, kill or irrigation (minimum 1).
pheno_reset	-	-	per year	-	Restarts the phenology of plants with a moisture_gro trigger in plants.plt.
cn_update	absval, abschg, pctchg or relchg	value	per year	-	Changes the curve number for moisture condition II: set to the value, add the value, change by the value in percent, or multiply by (1 + value). The result is kept between 35 and 95.

act_typ	option	const	const2	fp	Effect
tiledep_control	-	drain depth, mm	per year	-	Sets the tile drain depth for drainage water management.
impound_on	-	-	-	name in wetland.wet	Turns on impounded water (paddy or wetland) on the HRU.
impound_off	-	-	-	-	Turns off impounded water on the HRU.
weir_height	wet, or other	height, mm	-	-	With wet, sets the HRU paddy weir height and the principal and emergency volumes from it. Otherwise sets the weir height of reservoir obj_num.
puddle	name in puddle.ops	-	per year	-	Puddles a paddy (see management.sch). The action name must be the tillage operation in tillage.til used for the soil mixing.
lu_change	-	-	-	name in landuse.lum	Changes the HRU's land use and management and reinitialises plants and curve number. Logged in lu_change_out.txt.
snow_change	-	-	-	name in snow.sno	Switches the HRU to another snow parameter set. Logged in lu_change_out.txt.
hru_fr_update	new ls_unit.ele file	-	-	new rout_unit.ele file	Updates HRU area fractions. Logged in lu_change_out.txt.
p_factor	-	new USLE P	-	-	Sets the USLE P factor.
contour	-	-	-	-	Sets $P = 2.7\,S + 0.26$, with S the slope (m/m).
stripcrop	-	-	-	pasture or other	Sets $P = 2.32\,S + 0.36$, halved when fp is pasture.
terrace	-	-	-	sod_outlet or other	Sets $P = 1.4\,S + 0.13$, doubled when fp is sod_outlet.
tile_install	-	-	-	name in tiledrain.str	Installs tile drains (see *.str).
septic_install	-	-	-	name in septic.str	Installs a septic system.
fstrip_install	-	-	-	name in filterstrip.str	Installs a filter strip.

act_typ	option	const	const2	fp	Effect
grassww_install, grassww_uninstall	-	-	-	name in grassedww.str	Installs or removes a grassed waterway.
user_def_bmp	-	-	-	name in bmpuser.str	Applies user-defined BMP removal fractions.
grow_init, grow_end	-	-	-	-	Start and end of the growing season of an HRU-lte (**obj hlt**). **grow_end** computes the yield.
chan_change	-	cover	stream order	-	Sets the cover and stream order of the channel being checked.
herd	-	-	-	-	No effect in 62.0.1.

The `plant`, `harvest` and `harvest_kill` actions of the generic tables `pl_hv_summer1`, `pl_hv_winter1` and `pl_hv_summer2` take the plant name from `management.sch` instead of `option` (see `management.sch`).

Flow control and water allocation actions, used in `flo_con.dtl`:

act_typ	option	const	const2	Effect
`tileflo_control`	`flo_mm`, `max_mm` (at most `const` mm), `min_mm` (at least `const` mm), `all_flo`, `zero_flo`, `frac` (`const` × tile flow)	mm or fraction	-	Sets the tile flow sent from a saturated buffer source HRU to its receiving HRU.
`wallo_dmd`	`set_dmd`	demand, m^3	-	Sets the demand of a water allocation place of use.
`wallo_dmd`	`irr_dmd`	-	-	Adds the irrigation demand of place of use `obj_num`.
`wallo_dmd`	`fill_res`, `fill_wtow`, `fill_can`	target fraction of capacity	fraction of the shortfall per day	Demand to fill reservoir, water tower or canal `obj_num` to `const` times its capacity (for `fill_res`, `fp` `pvol` or `evol`).
`duty_fr`	`fr1` to `fr6`	fraction	-	Fraction of the demand taken from point of diversion 1 to 6 (or sent to return 1 to 6).
`duty_reset`	-	point of diversion number	new maximum withdrawal	Resets the withdrawal limit of a point of diversion of place of use `obj_num`.

See water allocation for how these demands and fractions are used.

Reservoir release actions

In `res_rel.dtl` the action type is `release` and `option` selects how the release is computed. The releases of all actions that run on a day are added together. A table with zero alternatives runs all its actions every day. V is the current volume, V_p the principal and V_e the emergency volume.

option	const	const2	fp	Release
`rate`	rate, m^3/s	-	-	constant rate
`rate_pct`	percent	-	-	`const` % of V_p per day
`inflo_rate`	maximum rate, m^3/s	change to inflow, m^3/s	-	the larger of inflow + `const2` and `const`
`inflo_frac`	fraction	-	-	`const` × inflow
`ab_emer`	-	-	-	all volume above V_e
`days`	days	multiplier	`null`, `pvol` or `evol`	$(V - V_t)$/const, with target $V_t = 0$, `const2` × V_p or `const2` × V_e
`dyrt`	days	percent	`null`, `pvol` or `evol`	$(V - V_t)$/const plus `const2` % of V_p, with $V_t = 0$, V_p or V_e
`dyrt1`	days	percent	`con1`, `con2` or `con3`	as `dyrt`, with the target taken from condition 5, 4 or 3 of the table
`inflo_targ`	condition number	-	-	inflow plus all volume above V_p × that condition's `lim_const`
`weir`	-	-	name in `weir.res`	weir equation

option	const	const2	fp	Release
meas	-	-	name of a recall object	measured outflow
natlake	inactive storage, fraction of V_p	release coefficient, 1/day	-	natural lake outflow (Döll et al., 2003)
nonirr-h06	α	-	-	Hanazaki et al. (2006) scheme for non-irrigation reservoirs
irr-h06	α	β	-	Hanazaki et al. (2006) scheme for irrigation reservoirs
hydrop	amplitude	phase	-	HYPE hydropower scheme (Arheimer et al., 2019)

See *.res for the reservoir side of these options.

Land use scenarios

Tables in scen_lu.dtl run only if they are listed in scen_dtl.upd. That file has a title line, the number of entries, a header line, and one row per entry with three columns: max_hits, typ and the name of a table in scen_lu.dtl. Each listed table is checked for every HRU every day, before the HRUs are simulated. The usual actions are lu_change and snow_change.

Example

refdata/Ames_sub1/lum.dtl in the SWAT+ repository contains two tables:

```
lum.dtl: yearly irrigation limit example for Ames dataset
2

name                      conds      alts      acts       !limit yearly irrigation to 500 mm using irrigate
irr_year_irr                 4         1         1
var                        obj   obj_num          lim_var            lim_op  lim_const      alt1
irr_year                   hru         0             null                 -   500.00000         <
w_stress                   hru         0             null                 -   0.80000           <
phu_plant                  hru         0          phu_mat                 -     0.20000         >
leaf_area                  hru         0          null                    -     0.50000         >
act_typ                    obj   obj_num             name            option    const        const2
irrigate                   hru         0         sprinkler     sprinkler_ilm    25.00000   20.00000  unli
```

Table irr_year_irr has four conditions, one alternative and one action. The alternative is satisfied on a day when the irrigation applied so far is below 500 mm, the average water stress factor is below 0.8, the crop has accumulated more than 20% of its heat units, and the leaf area index is above 0.5. On such a day the HRU is irrigated with 25 mm using the sprinkler_ilm entry of irr.ops, at most 20 times a year. obj is hru, so the water comes from an unlimited source. The table is used by naming irr_year_irr as an automatic operation in management.sch. In 62.0.1 the irrigation total behind irr_year is not reset each year, so once 500 mm has been applied over the whole run, the table stops irrigating for the rest of the simulation.

A reservoir table from refdata/Osu_1hru/res_rel.dtl, followed by the blank line that res_rel.dtl needs:

```
name                      conds      alts            acts
paddy_auto_rel               2         2               2
var                        obj   obj_num          lim_var            lim_op     lim_const      alt1   alt
wet_depth                  hru         0          weir_hgt                 -       0.00000         >
wet_depth                  hru         0             null                  -       2.00000         >
act_typ                    obj   obj_num             name            option         const        const2
release                    hru         0       above_emer              days       1.00000       0.00000
release                    hru         0       below_emer              weir      10.00000       0.00000
```

Both alternatives need a water depth above 2 mm, and the first also above 0 mm. When the first alternative is satisfied, the paddy releases its whole volume within one day (`days` with `const` 1 and target 0) and also releases over the weir `weir1`.

Related files

- `management.sch`: lists the `lum.dtl` tables each HRU uses.
- `*.ops`, `tillage.til`, `fertilizer.frt`, `landuse.lum`, `snow.sno`, `*.str`: databases named by actions.
- `*.res` and `*.wet`: name the release tables in `res_rel.dtl`.
- Water allocation and manure allocation: name tables in `lum.dtl` and `flo_con.dtl`.
- `file.cio`: names the four files on its `decision_table` line.

Sources

- Readers: `dtbl_lum_read.f90`, `dtbl_res_read.f90`, `dtbl_scen_read.f90`, `dtbl_flocon_read.f90`, `cal_cond_read.f90` (`scen_dtl.upd`)
- Type definitions: `conditional_module.f90`
- Conditions: `conditions.f90`, `cond_real.f90`, `cond_integer.f90`
- Actions: `actions.f90`; reservoir releases: `res_hydro.f90`
- Linking tables to HRUs: `proc_cond.f90`, `hru_dtbl_actions_init.f90`
- Callers: `hru_control.f90`, `time_control.f90`, `res_control.f90`, `wallo_start_day.f90`, `mallo_control.f90`
- File name slots: `type input_condition` in `input_file_module.f90`

Column	Fortran variable
condition `var`, `obj`, `obj_num`, `lim_var`, `lim_op`, `lim_const`	`d_tbl%cond(ic)%var`, `%ob`, `%ob_num`, `%lim_var`, `%lim_op`, `%lim_const`
`alt1` …	`d_tbl%alt(ic, ialt)`
`prob_unif` probability (`obj` column)	`d_tbl%frac_app`
action `act_typ`	`d_tbl%act(iac)%typ`
action `obj`, `obj_num`, `name`, `option`, `const`, `const2`	`d_tbl%act(iac)%ob`, `%ob_num`, `%name`, `%option`, `%const`, `%const2`
action `fp`	`d_tbl%act(iac)%file_pointer`
`outcome1` …	`d_tbl%act_outcomes(iac, ialt)`